PetPooja AI — Project Documentation

AI-Powered Revenue Intelligence & Voice Copilot for Restaurants

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1. Introduction

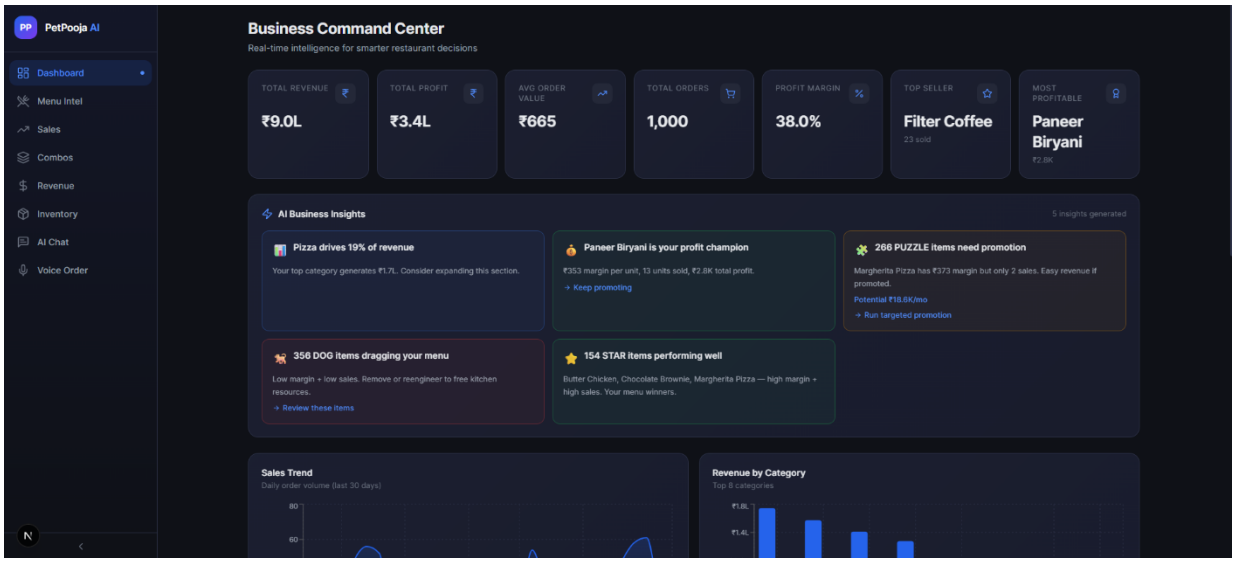
PetPooja AI is an **AI-powered restaurant intelligence platform** designed to bridge the gap between raw transactional data and real-world business decision-making.

Modern restaurants generate large volumes of Point-of-Sale (PoS) data every day. This data includes orders, item performance, pricing, customer behavior, and operational patterns. However, in most cases, this data is **either underutilized or used only for basic reporting**, rather than driving meaningful decisions.

As a result, restaurants face:

- Missed revenue opportunities
- Inefficient menu strategies
- Poor demand forecasting
- Operational inefficiencies

PetPooja AI addresses this problem by transforming static data into a **dynamic, AI-driven decision system**. By combining **Machine Learning, Data Analytics, and Voice AI**, the platform enables restaurants to not just understand their data — but actively **act on it in real time**.



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2. Project Vision

The core vision behind PetPooja AI is:

To build a system that acts as a real-time revenue copilot for restaurants.

Traditional analytics platforms are reactive — they tell you what happened.
PetPooja AI is proactive — it tells you what *will happen* and what you *should do next*.

The system is designed to:

- Predict demand before it occurs
- Identify revenue opportunities automatically
- Suggest actions that improve profitability
- Automate interactions with customers

Instead of relying on manual intuition, restaurants can now rely on **data-backed intelligence**, making operations more efficient and scalable.

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3. Problem Statement

The restaurant industry faces three major structural challenges:

3.1 Inefficient Use of Data

Although PoS systems collect detailed transactional data, most restaurants do not use it effectively. Decisions related to pricing, menu design, and promotions are often based on intuition rather than data.

3.2 Revenue Leakage

A significant portion of potential revenue is lost due to:

- Missed upselling opportunities
- Poor pricing strategies
- Lack of combo optimization

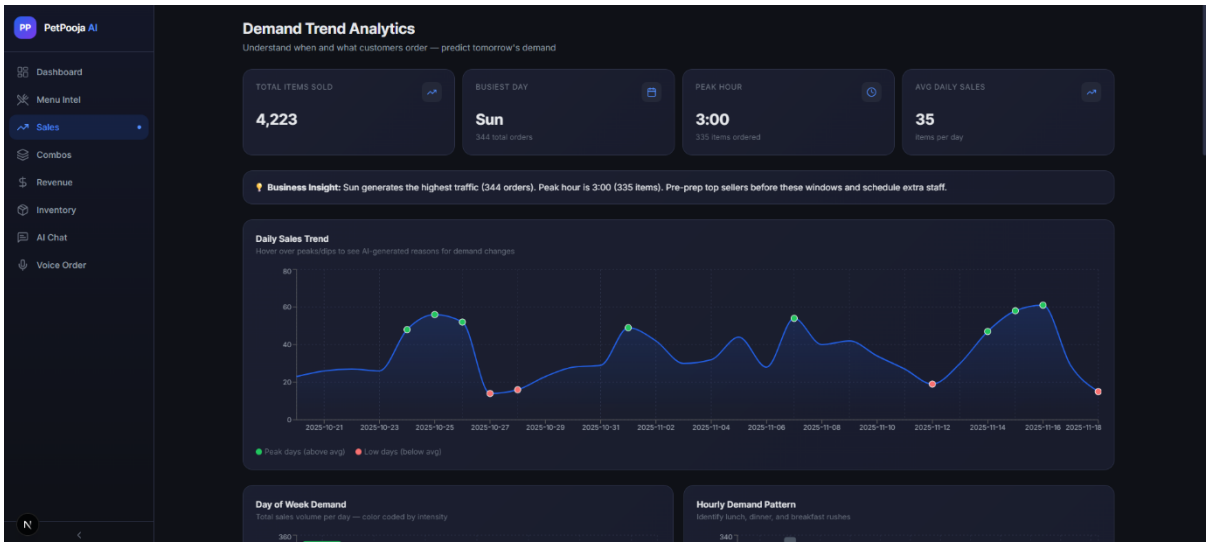
High-margin items may remain under-promoted, while low-margin items dominate sales.

3.3 Manual and Error-Prone Operations

A large percentage of orders are still handled manually through phone calls. This leads to:

- Miscommunication
- Order errors
- Increased dependency on staff
- Missed opportunities for upselling

These challenges highlight the need for an **intelligent, automated, and predictive system** — which PetPooja AI aims to provide.



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4. System Architecture

PetPooja AI is built using a **modular and scalable architecture**, where different components handle specific aspects of intelligence and automation.

At the core of the system lies a **data processing layer** that aggregates and structures PoS data. This data is then fed into multiple machine learning models, each responsible for a different business function.

This architecture ensures:

- Separation of concerns
- Easy scalability
- Real-time responsiveness

- Seamless integration between modules

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5. Core System Modules

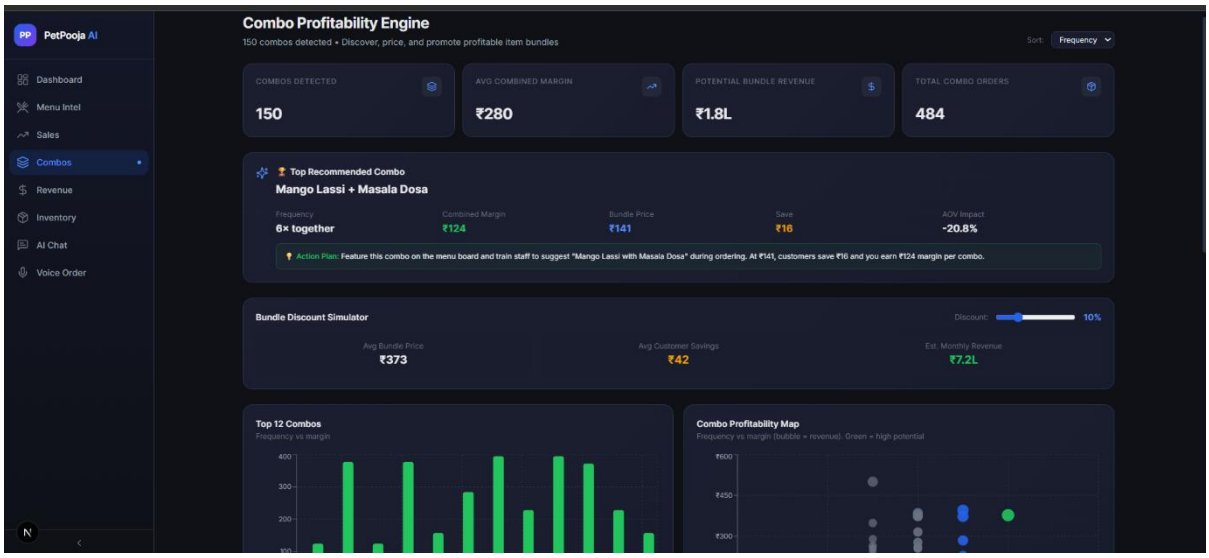
5.1 Executive Dashboard

The dashboard acts as the central interface for restaurant owners and managers. It provides a **comprehensive overview of business performance**, combining both historical data and predictive insights.

Key features include:

- Revenue KPIs
- Sales trends and forecasts
- Hourly demand heatmaps
- Category-level performance

Unlike traditional dashboards, this system integrates predictions, enabling **forward-looking decision-making**.



5.2 Menu Intelligence Engine

The Menu Intelligence Engine evaluates the performance of each menu item using a combination of **sales data and profit margins**.

Each item is classified into one of four categories:

- STAR (high sales, high profit)
- PLOWHORSE (high sales, low profit)
- PUZZLE (low sales, high profit)

- DOG (low sales, low profit)

This classification allows restaurant owners to:

- Promote high-potential items
- Optimize pricing strategies
- Remove underperforming items

The system transforms raw menu data into **clear strategic insights**.

5.3 Demand & Sales Engine

The Demand Engine is responsible for forecasting future demand based on historical patterns.

It predicts:

- Hourly demand fluctuations
- Daily and weekly trends
- Peak demand periods

This enables restaurants to:

- Optimize staffing
- Manage inventory efficiently
- Prepare for high-demand periods

By anticipating demand, businesses can shift from reactive to **proactive operations**.

5.4 Combo Recommendation Engine

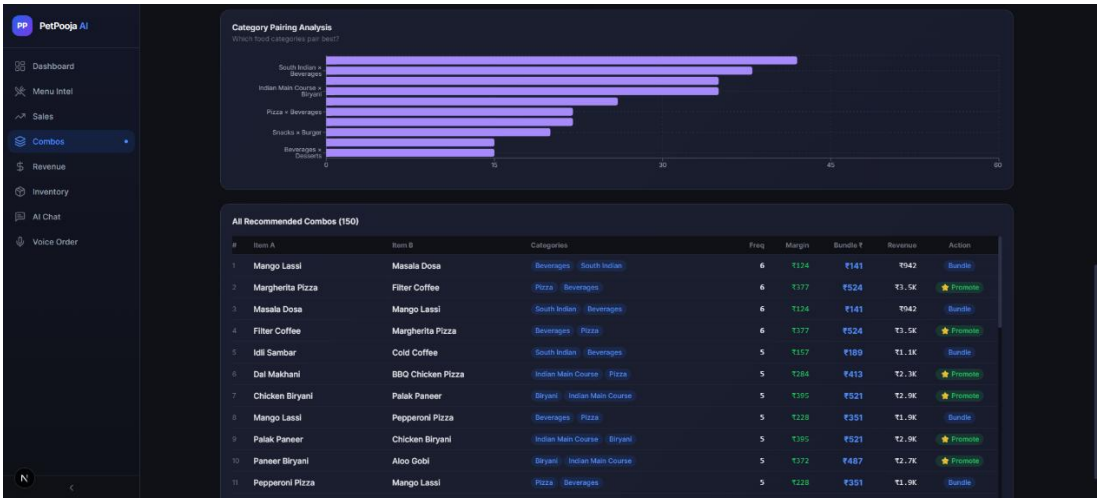
The Combo Engine identifies patterns in customer ordering behavior using association analysis.

It detects:

- Frequently purchased item combinations
- High-performing bundles
- Opportunities to increase order value

These insights are used to:

- Generate dynamic combos
- Suggest add-ons during ordering
- Improve Average Order Value (AOV)



5.5 Revenue Optimization Engine

This module focuses on maximizing profitability through **data-driven pricing strategies**.

It allows restaurants to:

- Simulate pricing changes
- Analyze the impact of discounts
- Identify high-profit opportunities

Instead of guessing prices, businesses can now rely on **predictive revenue modeling**.

5.6 Inventory Intelligence

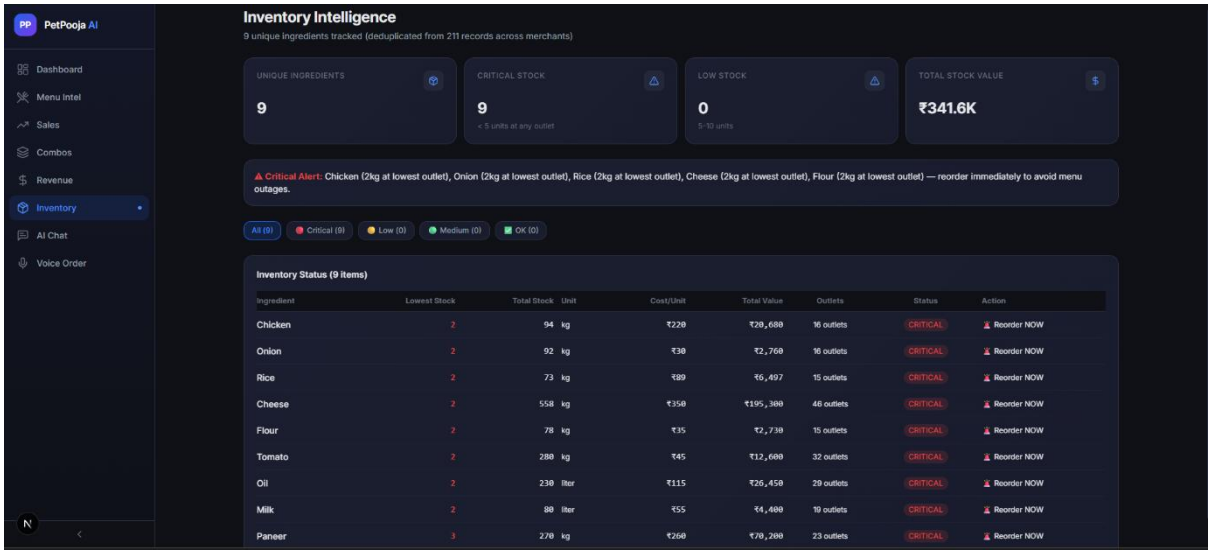
Inventory management is directly linked to demand forecasting.

This module predicts:

- When stock will run out
- What items need reordering
- Optimal inventory levels

This helps prevent:

- Stock shortages
- Overstocking
- Wastage



5.7 AI Voice Ordering Copilot

One of the most innovative components of the system is the **AI Voice Ordering Copilot**.

It automates the entire order-taking process by:

1. Converting speech to text
2. Extracting order intent
3. Mapping items to the menu
4. Validating modifiers and customizations
5. Suggesting upsell options
6. Generating structured orders
7. Sending orders directly to the PoS system

This reduces dependency on manual staff and ensures:

- Faster order processing

- Higher accuracy
- Increased upselling

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6. Machine Learning Architecture

The intelligence of PetPooja AI is powered by **four core machine learning models**, each designed to solve a specific business problem.

6.1 Demand Prediction Model

This is the **central model** of the system.

It uses historical data such as time, item category, and past sales to predict future demand.

The model is implemented using:

- Random Forest
- XGBoost

Its predictions are used across multiple modules, including:

- Dashboard forecasting
- Inventory planning
- Sales analysis

6.2 Menu Classification Model

This model categorizes menu items based on performance.

Initially implemented using rule-based logic, it provides:

- High interpretability
- Fast computation

It can be extended into a machine learning classification model for predictive categorization.

6.3 Revenue Optimization Model

This model uses regression techniques to estimate how changes in pricing or discounts affect revenue.

It enables:

- What-if analysis

- Profit simulation
- Strategic pricing decisions

6.4 Combo Recommendation Model

This model uses the Apriori algorithm to identify relationships between items.

It helps generate:

- Dynamic combos
- Cross-selling opportunities
- Personalized suggestions

6.5 Inventory Forecasting

Derived from the demand model, this component predicts stock consumption patterns and helps in planning inventory accordingly.

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7. User Interface & Experience

The user interface is designed with a focus on **clarity, usability, and actionability**.

The system provides:

- Interactive dashboards
- Visual charts and insights
- Real-time recommendations
- Voice interaction interface

Each component is built to ensure that users can:

- Quickly understand insights
- Take immediate action
- Navigate seamlessly

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8. Technology Stack

Frontend

- Next.js
- TailwindCSS

Backend

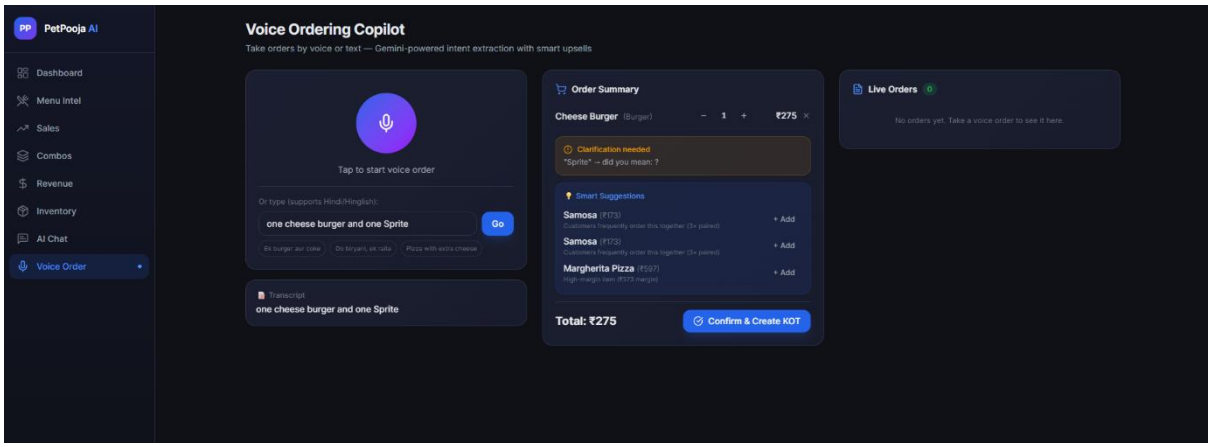
- Supabase (PostgreSQL + APIs)

Machine Learning

- Scikit-learn
- XGBoost
- Apriori Algorithm

AI & Voice

- LLM APIs
- Speech-to-Text
- Text-to-Speech



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9. Database Design

The system uses a structured relational database with analytical views.

Core Tables

- Merchant
- Menu
- Orders
- Order Items
- Inventory
- Recommendations
- Kitchen Orders

Analytical Views

- Item sales analysis
- Menu performance classification
- Combo detection
- Upsell candidates
- Inventory alerts

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10. Deployment

The application is deployed as a scalable web platform.

Steps include:

1. Hosting code on GitHub
2. Deploying frontend via Vercel
3. Managing backend through Supabase
4. Configuring environment variables

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11. Challenges & Limitations

- Accuracy of voice recognition in noisy environments
- Dependence on high-quality PoS data
- Cold start problem for new restaurants
- Complexity in mapping voice input to menu items

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12. Conclusion

PetPooja AI demonstrates how modern AI technologies can transform traditional restaurant operations into **intelligent, automated, and data-driven systems**.

By integrating:

- Predictive analytics
- Machine learning
- Voice automation

the platform evolves beyond a dashboard into a **complete decision intelligence system**.